James Horine

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PROFESSIONAL EXPERIENCE

BAYER – Data Science & AI **April 2020 – Current**

Applied Science Lead, AI Engineering & Solution Architecture

Solution architect and strategic AI partner to Bayer Crop Science business leaders, owning the path from use-case intake to production-ready agentic AI, GenAI, ML, and analytics solutions that inform decision-making across the division.

- **Agentic Platform Development:** Lead the design, solution architecture, development, and implementation team for agentic AI platforms supporting the AI portfolio. Owns the business case, multi-agent architecture, governance model, and co-developed evaluation patterns.
- **Solution Architecture & Rapid Prototyping:** Owned and defined solution architecture, evaluations, and success criteria for prototype agentic, AI, machine learning, and geospatial analytics solutions on the platform.
- **Proposal Management:** Run the intake, prioritization, and execution of high-impact AI initiatives, engaging business stakeholders to surface use cases and assessing each for business value, technical feasibility, and data readiness before investment.
- **Integration & Scaling:** Transition proven capabilities into the enterprise portfolio with platform and engineering teams, ensuring scalability, ownership, and fit with business processes.

Lead Data Scientist, Program Manager

Manager delivering solutions for global agricultural research, economics, and policy programs, owning technical execution, solution architecture, staffing, and scope against business objectives.

- **Data Science Leadership:** Led strategic initiatives in GenAI, forecasting, ML, and agricultural economics, setting technical direction and review standards across cross-functional teams.
- **Contract Evaluation:** Evaluated vendor and internal model solutions and architectures with stakeholders, aligning build, buy, and partner decisions with enterprise requirements.
- **Program Development:** Directed cross-functional programs, delivering a novel GenAI policy-risk solution, expanding the global envirotype model, and architecting the market share program.

Data Science Team Lead

Lead multi-disciplinary team, taking research from inception to production pipeline and owning the scientific and engineering roadmap for the Crop Disease Capability.

- **Machine Learning Innovation:** Developed patent-pending ML algorithms (US20240119542A1), improving predictive model accuracy by 50%.
- **Team Delivery:** Led the team that architected and delivered the scientific pipeline, accelerating delivery of model advances to stakeholders by 10x.
- **Professional Recognition:** Received the 2021 Bayer Top Performance Award for data science innovation and technical leadership.

UNITED AIRLINES – Operations Data Science **May 2019 – April 2020**

Senior Manager of Data Science

Leader of the operations data science team serving 10+ business areas, owning technical direction, solution architecture, staffing, and scope against business objectives.

- **Team Leadership:** Led delivery across Contact Centers, Technical Operations, Social, Load Planning, Safety, and Supply Chain, balancing priorities across competing business owners.
- **Operational Efficiency:** Led predictive modeling that reduced case closure time by 20% and lowered outbound reimbursement cost per case.
- **Technical Implementation:** Directed development of a baggage-prediction ML model, saving $150,000 in fuel overages.

SITO MOBILE – Data Team **February 2018 – November 2018**

Senior Data Scientist

Senior engineer and analytics developer on the client pipeline for ad-serve analytics and telemetry.

- **Analytics Innovation:** Developed the Brand Momentum Index, measuring foot-traffic dynamics across all locations for major brands.
- **Location Intelligence:** Architected proprietary geolocation analytics that distinguished unique visitors from unique visits, enabling customer journey analysis.
- **Technical Development:** Built Scala and Python petabyte scale production pipelines

GENERAL ASSEMBLY **June 2017 – October 2018**

Data Science Instructor

Instructor and mentor to career-changing practitioners in an industry data science curriculum.

- **Curriculum Leadership:** Updated and delivered curriculum covering Python, regression, machine learning, time-series forecasting, and NLP.
- **Student Success & Career Development:** Mentored on capstone projects and career positioning, including a graduate admitted to the University of Chicago's MS in Data Science program.
- **Technical Instruction & Mentorship:** Delivered hands-on instruction in Python and advanced analytics, providing individualized project feedback to mixed-experience cohorts.

THE MARKETING STORE **February 2017 – November 2017**

Senior Data Scientist

Senior AI/ML developer for global automotive, quick-service restaurant (QSR), and telecom clients.

- **Predictive Modeling:** Built ML recommendation systems that identified upsell opportunities for clients' digital and hybrid campaigns.
- **Behavior Simulation Modeling:** Built agent-based simulations of customer behavior from maintenance history and engagement data, characterizing upsell scenarios.
- **Engineering Excellence:** Established automated unit testing and CI/CD for data science pipelines, making model and engineering releases repeatable.

CAPGEMINI **April 2016 – February 2017**

Senior Consultant

Data science consultant for Fortune 500 clients in QSR, agriculture, and heavy industry.

- **Client Strategy Leadership:** Shaped data science narratives and strategic frameworks for Fortune 500 clients, managing multiple concurrent PoC and full engagements.
- **Advanced Modeling Solutions:** Delivered sales forecasting models for major QSR clients, a failure-prediction model for heavy industrial tires, and ML models for crop disease.

- **Technical Innovation:** Developed a novel feature-ranking methodology, quantifying feature contribution to support ML model interpretation.

SPIRALEDGE **June 2014 – January 2016**

Applied Data Scientist / Machine Learning Researcher

Applied ML researcher on time-series and signal-processing products for sports performance, from physical product testing to research-grade data collection.

- **Algorithm Innovation:** Developed novel motion-tracking and analysis algorithms for competitive distance athletes.
- **Statistical Methodology:** Designed new methods for biomechanical motion analysis, advancing athlete performance measurement.
- **Performance Validation:** Established performance metrics and standards using time-series analysis and nonparametric classification.

Academic Experience

George Mason University **Jan 2026 – Current**

Adjunct Professor – Cyber Security Engineering

- **Course Development**: Develop and expand AI-agnostic curriculum for graduate cyber security engineering capstone practicum, guiding student teams to solution delivery.

Education

Penn State University **Anticipated completion 2027**

Doctor of Engineering, Systems for Spatial Econometrics

San Jose State University

Master of Science, Statistics

University of California, Santa Cruz

Bachelor of Arts, Mathematics

Open-Source Software Development

R Library Developer

- `useful-views`*:* Agentic framework for data projects to record the recipes that derive every table.
- `fastFeatures`*:* approximately correct feature selection in large feature spaces.
- `TidyAgronomy`*:* scalable modern data science ecosystem for agronomy.
- `Getting Started with Open Source`: Overcoming barriers in Open-Source projects.

Patents

Patent Pending - *US20240119542A1:* Systems and methods for assessing crop damaging factors associated with agronomic fields

Talks and Lectures

OSTEM, *Careers in Academic and Industrial Research* 2017 Panelist

Galvanize SF, *A Crash Course in Time Series* 2015 Speaker